

Absolute value inequalities



1 Consider the inequality $|x| \leq 5$.

- a Rewrite the inequality without absolute values.
- b Graph the solutions to the inequality on a number line.

2 Complete the following statement:

If $c > 0$, then $|u| < c$ is equivalent to $\text{ } < u < \text{ }.$

3 Rewrite the following inequalities without using absolute values:

- a $|x| < 2$
- b $|x| \geq 5$

4 Write the following inequalities as absolute value inequalities:

- a $-7 \leq x \leq 7$
- b $x < -9$ or $x > 9$
- c $-0.6 \leq 2x + 3 \leq 0.6$

5 Write an absolute value inequality that represents:

- a All real numbers x that are less than 8 units away from 0.
- b All real numbers x that are more than 8 units away from 0.
- c All real numbers x that are at least 2 units away from 8.
- d All real numbers x that are at most 5 units away from -2 .

6 Graph the number line representation of each of the following:

- a $|x| \leq 3$
- b $|x| \neq 2$
- c $|x| > 6$
- d $|x| \neq 6$
- e $|x| < 7$
- f $|x| = -7$
- g $|x| \leq 9$
- h $|x| = 9$
- i $|x| \geq 5$
- j $|x| > -5$

7 Which of the following statements is equivalent to $|2x - 2| > -5$?

- A $2x - 2 \leq -5, 2x - 2 \geq 5$
- B $2x - 2 = x, 2x - 2 = -x$
- C $-5 \leq 2x - 2 \leq 5$
- D $2x - 2 = 5, 2x - 2 = -5$
- E $(-\infty, \infty)$

8 Solve the following inequalities:



a $|x - 7| \geq 2$

b $|x - 5| > 11$

c $|x - 4| \leq 9$

d $|x + 2| < 6$

e $|x + 3| \leq 7$

f $|3x + 5| < 11$

g $|2x + 3| \leq 5$

h $2 + |x| \geq 9$

i $|x| - 6 > 9$

j $|11 - 2x| > 5$

k $|8 + 6x| > 4$

l $|x - 7| - 6 \geq 2$

m $|3x + 4| \geq x + 8$

Applications

- 9 Cell phone cases have dimension requirements to ensure the phone will fit properly in the case. The manufacturing engineer has written the specification that the new length, n , of the case can differ from the previous length, p , by only 0.04 centimeters or less. This is represented by the inequality:

$$|n - p| \leq 0.04$$

Find the limits of the new length of a cell phone case if the previous length was 18.9 centimeters.

- 10 If a coin is tossed 100 times we would expect approximately 50 of the outcomes to be heads. It can be demonstrated that a coin is unfair if h , the number of outcomes that result in heads, satisfies $\left| \frac{h - 50}{5} \right| \geq 1.645$.

a Solve the inequality.

b Complete the following sentence using integer values:

If a coin is tossed 100 times and the number of outcomes that result in heads is $\geq$ or less or $\leq$ or more, then the coin is unfair.

- 11 In a certain company, the measured thickness, m , of a helicopter blade must not differ from the standard, s , by more than 0.17 millimeters. The manufacturing engineer expresses this as the inequality $|m - s| \leq 0.17$.

Find the possible values of m if the standard, s , is 17.92 millimeters.